2.2.2 已知图 2.08 电路的参数及电流如图中所示，若 U_S 增加 2 V，则 I 为多少？

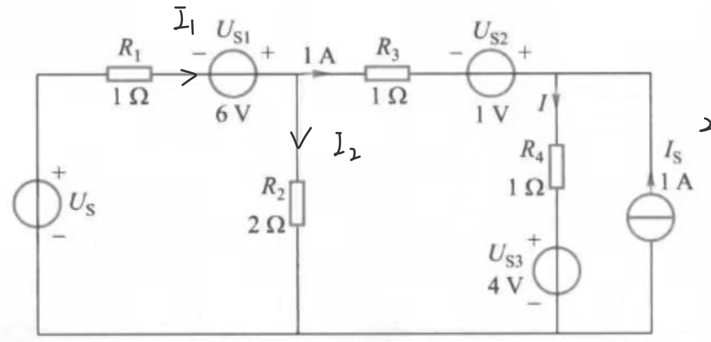
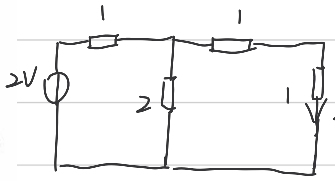


图 2.08 习题 2.2.2 电路

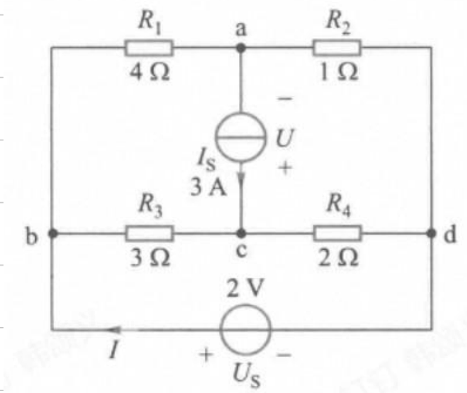
使用叠加原理，考虑 $U_s = 2V$.



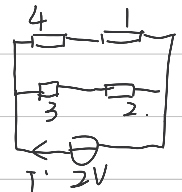
$I' = \frac{2}{1+\frac{2}{2}} \times \frac{1}{2} A = 0.5A$

$\therefore I = 1A + 1A + 0.5A = 2.5A$.

2.2.3 试用叠加定理求图 2.05 所示电路中流过 U_S 的电流 I，图中元件参数和题 2.1.5 相同。

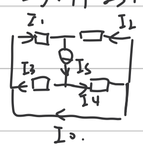


当只有 U_S 时.



$I' = \frac{2V}{5/2\Omega} = 0.8A$

当只有 I_S 时.



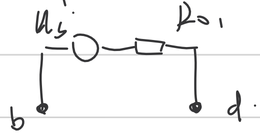
$\begin{cases} I_3 + I_4 = I_S \\ I_3 R_3 = I_4 R_4 \\ I_1 R_1 = I_2 R_2 \\ I_1 + I_2 = I_S \end{cases}$ $\begin{cases} I_1 = 0.6A \\ I_2 = 2.4A \\ I_3 = 1.2A \\ I_4 = 1.8A \end{cases}$

$\therefore I_0 = 0.6A - 1.2A = -0.6A$

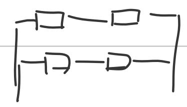
$\therefore I = I_0 + I' = 0.2A$

2.2.7 在图 2.05 所示电路中，将 U_S 以外的电路看成一个二端网络。试用等效电源定理求流过 U_S 的电流 I。

等效电源

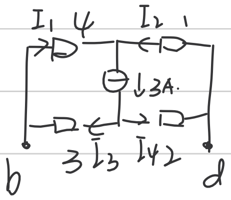


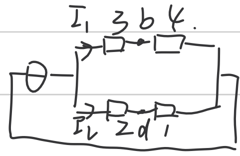
R_0:



$R_0 = \left(\frac{1}{5} + \frac{1}{5}\right)^{-1} = 2.5\Omega$.

U_S'.





$\begin{cases} 7I_1 = 3I_2 \\ I_1 + I_2 = 3A \end{cases}$ $I_1 = 0.9A$ $I_2 = 2.1A$.

$\therefore U = 4 \times 0.9V - 2.1V = 1.5V$

$\therefore I = \frac{U_S - U_S'}{R_0} = \frac{2 - 1.5}{2.5} A = 0.2A$.

2.2.8　如图 2.12 所示以 a、b 为端点的二端网络中，含有电流控制电流源 $20I_2$。试求：(1) 当 $R_3=10\ \text{k}\Omega$ 时的开路电压 U_{OC} 和等效电阻 R_0；(2) 当 $R_3=\infty$ 时的开路电压 U_{OC} 和等效电阻 R_0。

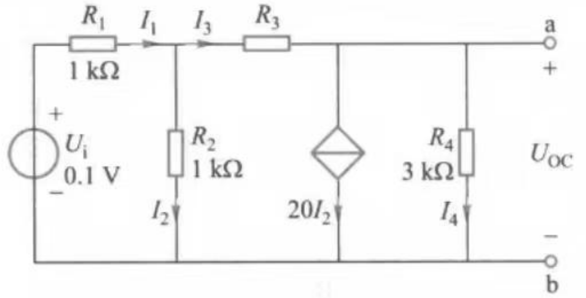


图 2.12　习题 2.2.8 电路

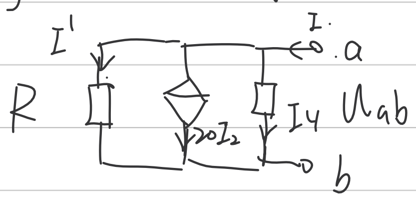
(1)

$$\begin{cases} I_3 = 20I_2 + I_4 \\ I_1 = I_2 + I_3 \\ U_i - I_1R_1 - I_3R_3 - I_4R_4 = 0 \\ U_i - I_1R_1 - I_2R_2 = 0 \end{cases}$$

$$\begin{cases} I_1 = 0.086\ \text{mA} \\ I_2 = 0.01494\ \text{mA} \\ I_3 = 0.07011\ \text{mA} \\ I_4 = -0.12287\ \text{mA} \end{cases}$$

$\therefore U_{OC} = I_4R_4 = -0.6861\text{V}$

将 U_i 置零，可简化电路：



由分流关系得 $I' = 2I_2$

$$\begin{cases} I = I_4 + 20I_2 + 2I_2 \\ I'R = I_4R_4 \end{cases}$$

$R_4 = \frac{U}{I_4}$

$R_0 = \frac{U}{I} = \frac{21}{29}\Omega$

$\approx 0.7241\Omega$

$R = R_3 + \frac{R_1R_2}{R_1+R_2} = 10.5\text{k}\Omega$

(2)　当 $R_3 = \infty$，即视为开路.

$I_2 = U_i/(R_1+R_2) = 0.05\ \text{mA}$.　　$U_{OC} = I_4R_4 = R_4\times(-20I_2)$

$= -3\text{V}$

2.2.9　如图 2.13 所示电路，$A_0=5\ 500$，试求：(1) a、b 两端左侧的开路电压 U_{OC} 和等效电阻 R_0；(2) a、b 右侧接入电阻 $R_L=2\ \text{k}\Omega$ 时，R_L 两端的电压 U_L。

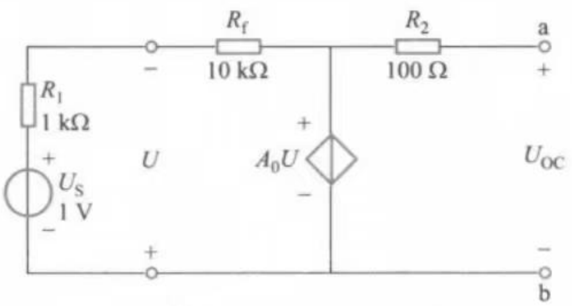


图 2.13　习题 2.2.9 电路

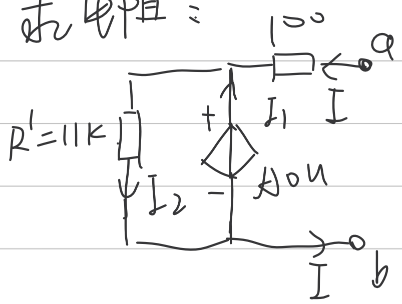
(1)　求开路电压：

$$\begin{cases} U_S - I(R_1+R_f) - A_0U = 0 \\ U = -(U_S - IR_1) \end{cases}$$

$\therefore I = \frac{U_S(A_0+1)}{R_1(A_0+1)+R_f}$

$\therefore U_{OC} = U_S - I(R_1+R_2) = -9.98\text{V}$

求电阻：



求a、b端短路电流

$$I_{短} = \frac{A_0 u}{R_2}$$

$$\therefore R_0 = \frac{u_{oc}}{I_{短}} = \frac{u_{oc} R_2}{A_0 u} = R_2 = 100\Omega$$

(2) $u_L = \frac{u_{oc} R_L}{R_0 + R_L} = \frac{-9.98}{100+2k} \times 2k\Omega = -9.5V$